# Andi Scarola

El Paso, Texas
LinkedIn | GitHub | Google Scholar | Portfolio

## Education

**The University of Texas at El Paso (UTEP)** — **El Paso, Texas**
*Master of Science in Artificial Intelligence* — GPA: 3.66 | **May 2027**
*Bachelor of Science in Computer Science; Minor: Mathematics* — GPA: 3.39 | **May 2026**

## Experience

**Vantage Innovation** — **May 2026 – Present**
***AI Engineer***

- Designed a **multi-agent execution harness** to coordinate specialized agents across complex workflows.

**W.M. Keck Center for 3D Innovation at UTEP** — **El Paso, Texas**
***Ceramics Team*** *(ML/Computer Vision Engineer)* — ***September 2025 – Present***

- Led a **U-Net**–based computer vision pipeline for fiber segmentation across **800**+ ceramic micrograph tiles.
- Implemented structure-tensor analysis and corrected density bias to quantify fiber alignment and coverage.

***Flow Team*** *(Lead Software Developer)* — ***May 2025 – September 2025***

- Architected a Flask dashboard for **3** airflow tools and optimized Python pipelines, cutting runtimes by up to **85%**.

**OSU NSF AI-EDGE Institute** — **Columbus, Ohio**
***Generative AI Researcher*** — ***May 2024 – July 2024***

- Built ASL recognition for **50**+ gestures with **MediaPipe**, **PyTorch**, and diffusion-based data augmentation.

**UTEP NSF Coding Like a Data Miner (CLDM)** — **El Paso, Texas**
***Research Assistant*** — ***October 2023 – October 2024***

- Co-authored **2** conference publications and mentored **20**+ students in **Python** through code reviews and debugging.

## Projects

**BorderETA** | *React Native, Python, FastAPI, Supabase, scikit-learn* — **April 2026 – Present**

- Shipped a border-crossing app for **2,000** users, fusing **8 real-time sources** into crossing estimates and routing.
- Trained quantile boosted trees on **58,016** border and **29,437** road-delay records with temporal holdouts.

**Bidwire** | *TypeScript, Claude API, Zod, Supabase, Next.js* — **September 2026**

- Orchestrated a **5**-stage contract-intelligence pipeline with Claude extraction, Zod validation, enrichment, and delivery.

**Clara (Elder Care)** | *TypeScript, Claude API, PostgreSQL, Plaid* — **October 2026**

- Engineered a bilingual **multi-agent elder-care system** for financial monitoring, document analysis, and support.
- Coordinated agents through **shared memory**, durable jobs, and consent checks, keeping financial calculations in code.

**Appraisal OS** | *TypeScript, Python, PostgreSQL, Zod, Next.js* — **May 2026 – Present**

- Cut appraisal workflow time from **4 hours** to **under 30 minutes**, including on-site photography of the home.
- Hardened PostgreSQL workers with lease fencing, **atomic artifact publication**, and revision-checked evidence review.

**ReconAIssance** | *YOLOv8, TensorRT, MAVSDK, PX4, Jetson Orin Nano* — **October 2025 – May 2026**

- Optimized YOLOv8 models trained on **10k**+ images with TensorRT, sustaining **30 FPS** inference on Jetson Orin Nano.
- Integrated **closed-loop vision and flight control** on a drone with no API, enabling autonomous repositioning.

## Technical Skills

**Languages:** Python, TypeScript, JavaScript, SQL, Java, C

**Technologies:** PyTorch, scikit-learn, OpenCV, FastAPI, React, PostgreSQL, Supabase, Docker, Cloudflare Workers

## Publications & Honors

*Cultural Relevance for Epistemic Practice in High School Computational Data Mining*. IEEE FIE, 2024.
*Sandboxing in Data Science: An Exploration of Youth Learning...* ICLS, 2024.

SFS Scholarship (2026) | 1881 Accelerator (2026)

**2nd Place:** Clara (2026), ReconAIssance (2025), Fight Coach (2025), DUI Risk Radar (2024).